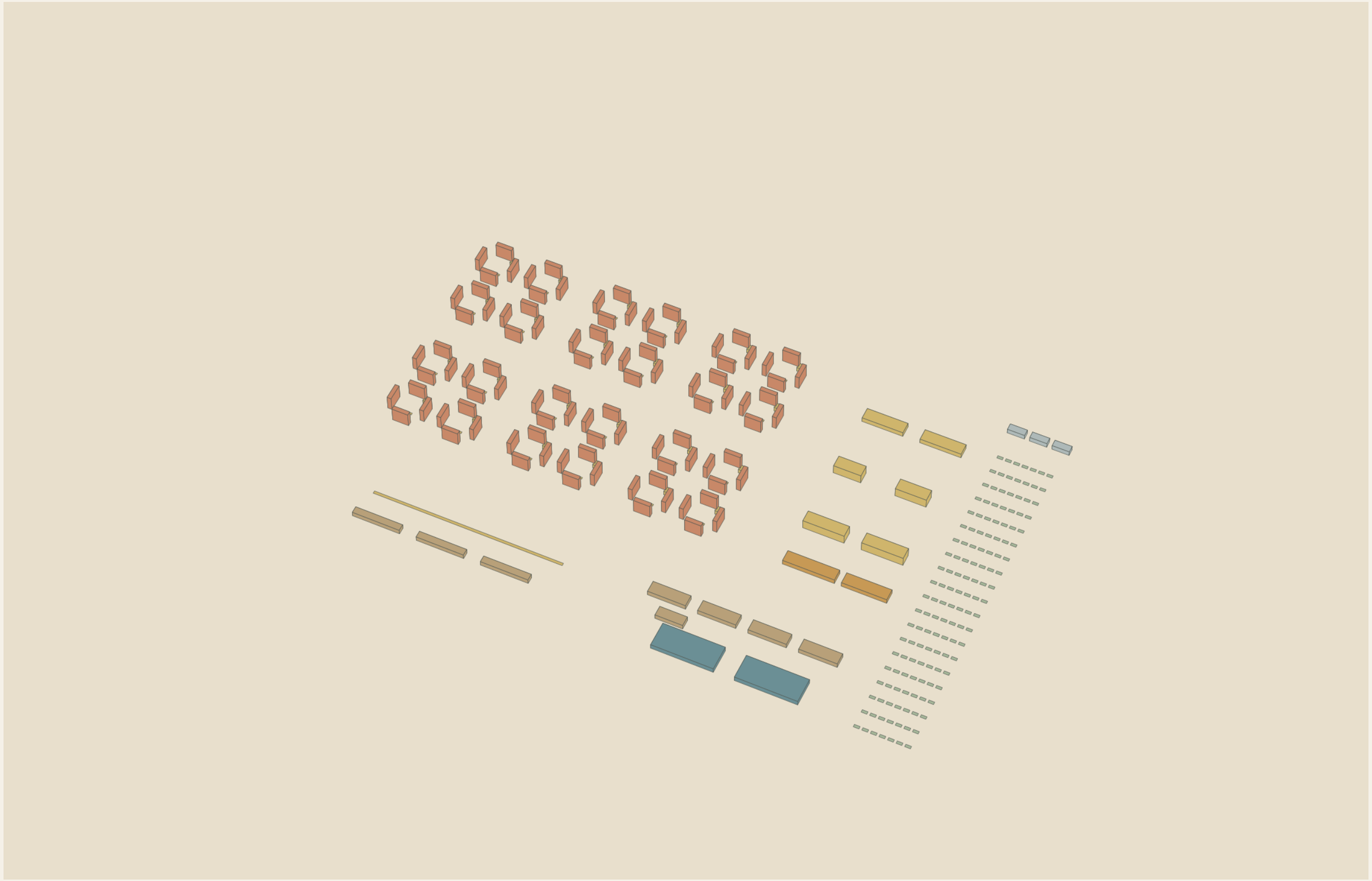
Geometry and method audit

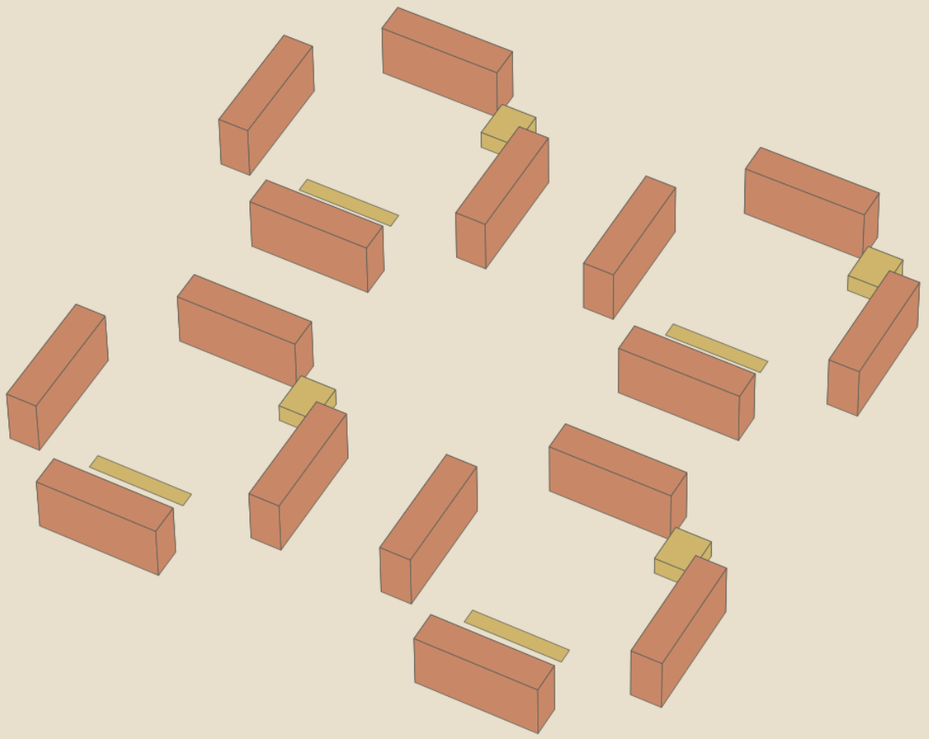
source_commit	9e504de
source_boundary_revision	2026-10-27-seam-closure
phase_one_planar_km2	2.770339362257745
housing_buildings	96
dwellings	1440
residents_scenario	3600
jobs_scenario	1500
interfaces	33
buildings	306
people_path_building_conflict_m	0.0
building_fixed_overlap_m2	0.0
building_channel_screen_overlap_m2	0.0
building_flood_screen_overlap_m2	0.0
land_partition_gap_m2	7.889616587886724e-07

Planar areas and geodesic published metrics have different methods. Flood and channel tests use issued preliminary screening. A/B/C IDs follow the current brief. Automatic file, geometry and numerical checks are separate from independent design review; no design approval is claimed.

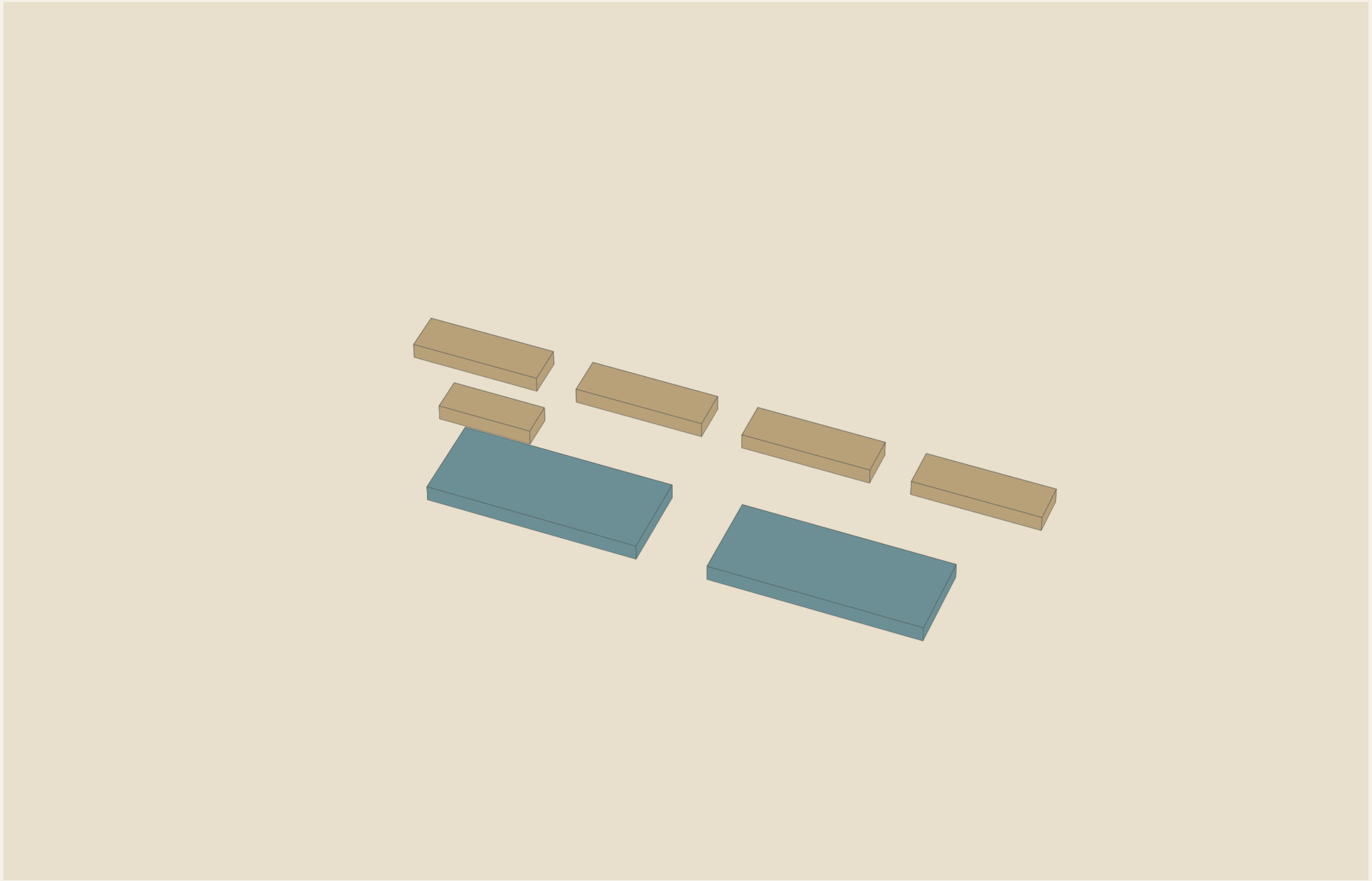
district / original geometry control



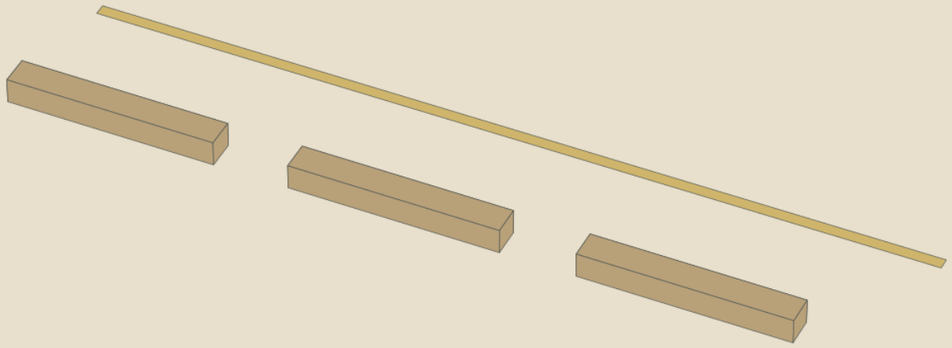
A01 / original geometry control



B03 / original geometry control



C02 / original geometry control



Declared assumptions

A-HOMES / Housing and services
1440 homes; 2.5 people/home; 96 wings, 15 homes/wing; 25% protected rental target; primary 360, secondary 240, childcare120 places
Original capacity programme. Household types and actual demand need survey. Of 1000 students, 600 are school-age,120 preschool,280 tertiary/apprentice scenario.

A-ENERGY / Electricity, cooling and backup
8 MW peak / 6 MW mean IT; PUE1.30; WUE0.015 L/kWh; 20 MW PV; 180 MWh battery; 20 MW grid; 72 MWh protected usable reserve; 95% charge/discharge efficiency
Explicit project scenario. Synthetic sun and periodic dust episodes in tools/energy_model.py; no utility confirmation. 16MW rooftop +4MW utility-yard canopies;220W/m2 module density;approximately90910m2 total module area;roof structural,fire,yield and interconnection verification required. Individual roof modules are not shown in all concept images.

A-WATER / Gross use and reuse
891748.4 m3/year total. Fresh abstraction751748.4; distributionloss40000; deliveredfresh711748.4; internalreuse180000
Domestic3600x140Lx365; IT52560000kWh x0.015L/kWh. Agriculture650000 includes fixed-pilot placeholder600000 and new50000; landscape12000;industry45000. No verified water rights. Fixed living-quarter demand unknown and additional.

A-SHORT / 30% reduction
267524.52 m3/year reduction:12000 landscape+35000 industry+180000 agriculture+40524.52 domestic/public
Residual use624223.88. Domestic143435.48=109.16 L/person/day; pilot allocation cuts require operator agreement and crop/animal welfare verification; otherwise reduce occupancy or secure emergency supply.

A-HEAT / Waste heat
2628 MWh/year usable heat, 5% of52560MWh IT electricity
Named receiver: cooperative drying/processing workshops; temperature, losses, seasonal demand and contracts unknown.

A-TERRAIN / Siting and drainage
Mean slope2.64% coarse footprint screening;150m channel research retreat;35m stopping distance outside fixed pilot
Scenario basis only, not surveyed banks, flood boundary, rights or accessible grading.

A-GROWTH / Regional capacity
First district3600; two later service-town scenarios1800each; conditional regional total7200 new residents
Growth not promised. Independent services and audited resource margin20%,occupancy80%,funded access required; actual layouts for later towns remain to be designed.

A-OPERATIONS / Traffic, lighting and services
10 km/h shared neighbourhood loops; machine stop at public crossings;2200K shielded lighting;1MW essential-service load
Concept assumptions pending traffic, lighting, seismic, fire and load studies. No facial recognition or household movement tracking.

A-GOVERNANCE / Asset continuity
Proposed civic trust, independent operator leases, maintenance/service reserve and resident council
Institutional design only. Legal form, reserve amount and affordability remain to be negotiated.

Source and boundary record

S-SNAPSHOT / XCITY issued competition site package
https://github.com/XcityUS/xcity-ai-competition/tree/9e504de/brief/site-package
Boundary, fixed pilots, terrain and drainage screening, regional roads. Not cadastral certification, utility rights or hydraulic approval.

Hourly energy model and resource accounts

fully_matched_hours_pct: 29.828767123287673

clean_energy_share_pct: 54.380216713320124

total_mwh: 78840.0000000067

grid_import_mwh: 35966.63714321495

peak_grid_mw: 11.799999999999999

limits: Synthetic hourly PV profile, not measured climate; starting battery at protected reserve, no clean credit from grid

Full 8,760-row model: report/hourly-energy.csv. Gross-use water account: 891,748.4 m3/year. Reuse: 180,000. Delivered fresh: 711,748.4. Distribution loss: 40,000. Required abstraction scenario: 751,748.4. All supply rights unverified; the fixed living-quarter demand is additional and unknown.